From TCV to TRUST: Adaptation of the MEQ Framework for Equilibrium Reconstruction and Full-Pulse Simulation

Academic Year 2025 - 2026

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THE MEQ FRAMEWORK

A numerical toolbox for axisymmetric tokamak equilibrium modelling

FBT	LIUQE	FGS	FGE
Inverse equilibrium	**Equilibrium reconstruction**	**Static forward equilibrium**	**Time-dependent forward simulation**

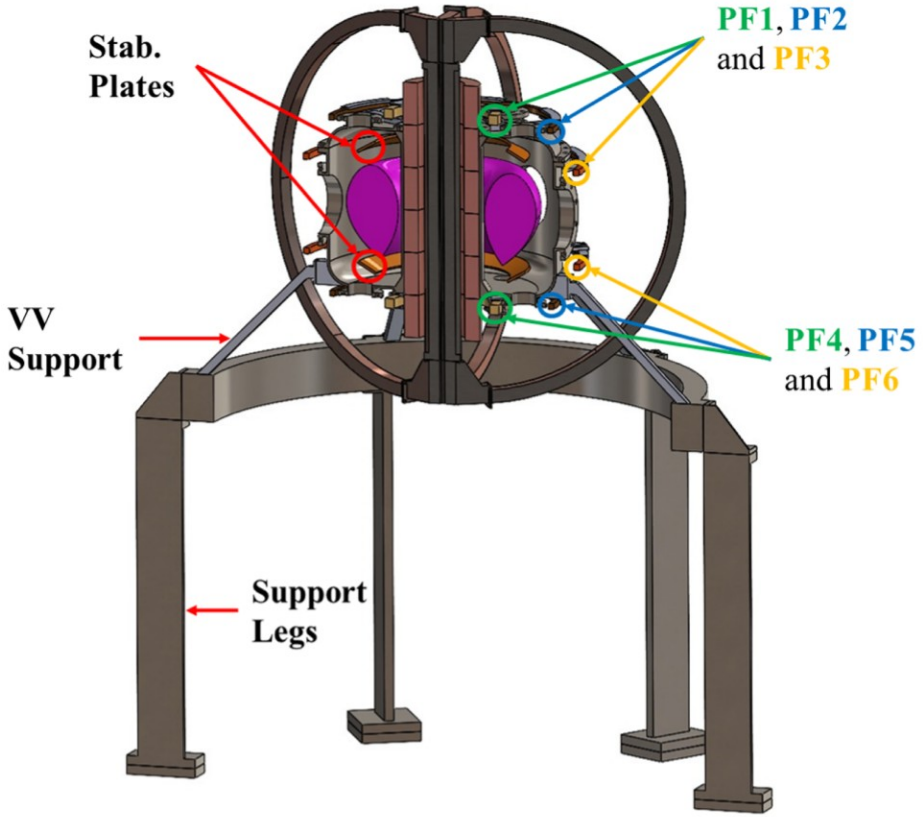
Same equilibrium physics – different computational problems

ADAPTING MEQ TO TRUST

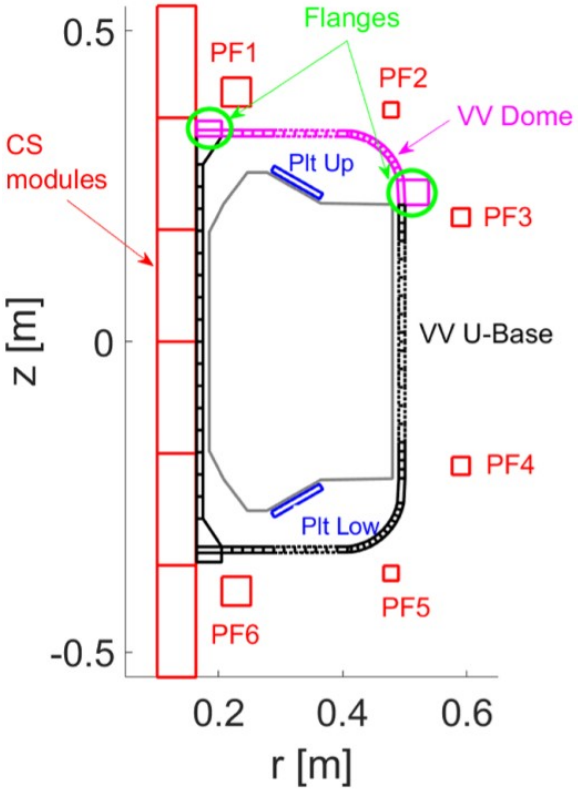
	GENERIC FUNCTIONS Shared across machines	MACHINE-SPECIFIC FUNCTIONS Defined for each device
PARAMETERS	Complete and check model settings	Define device-specific settings
GEOMETRY	Assemble the electromagnetic model	Describe conductors, wall and diagnostics
INPUT DATA	Complete and check solver inputs	Supply scenario data and measurements

Generic MEQ core retained; machine-specific functions adapted to TRUST

THE TRUST TOKAMAK



3D conceptual design

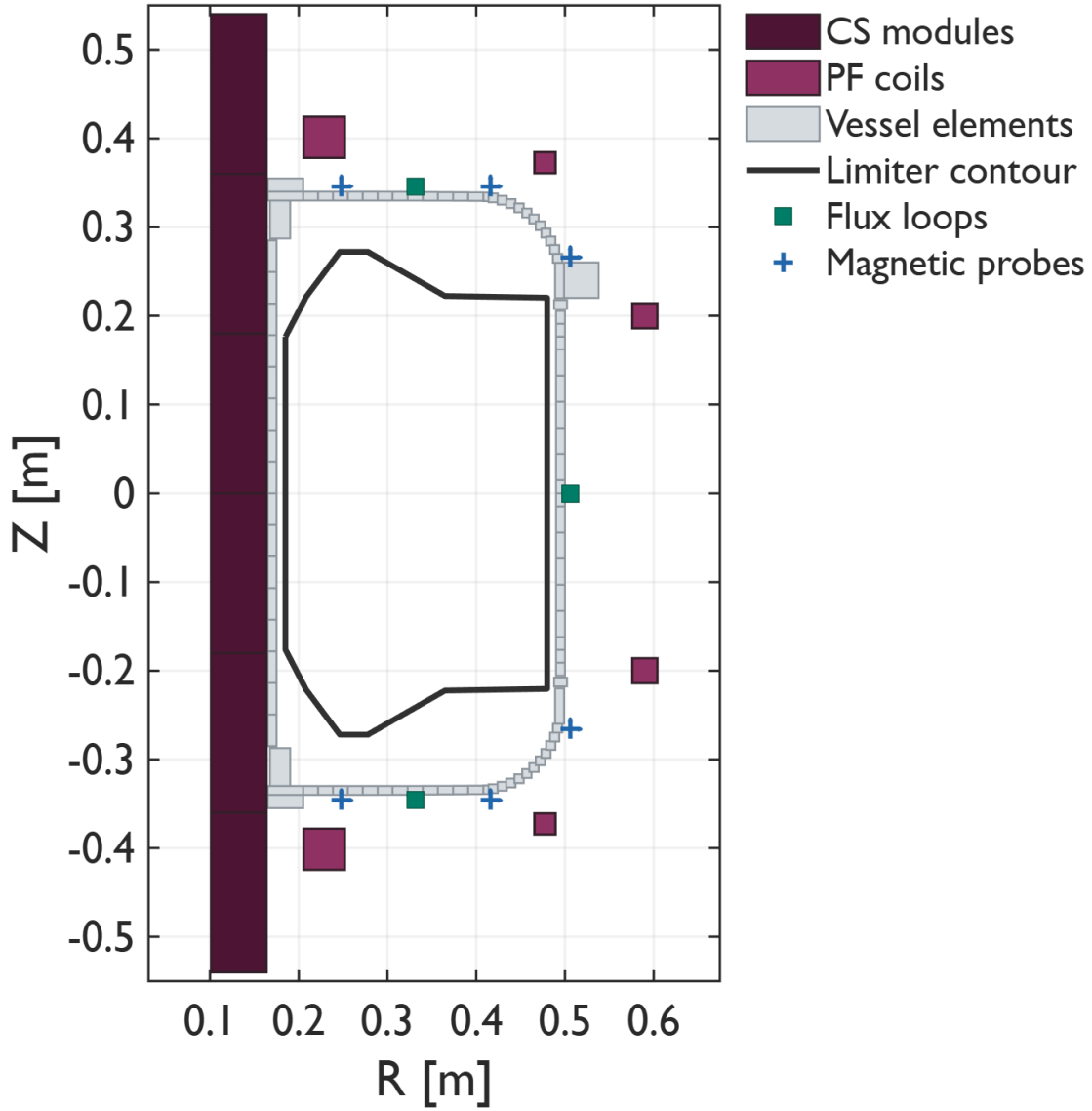


Poloidal cross-section

Reference design parameters

Major radius	Toroidal field	Plasma current
≈ 0.30 m	**≈ 0.8 T**	**110 kA**

THE TRUST-MEQ MACHINE MODEL



ACTIVE CIRCUITS

12 circuits: 6 CS modules + 6 PF coils

PASSIVE VESSEL

93 axisymmetric toroidal elements

MAGNETIC DIAGNOSTICS

Default layout shown

3 flux loops + **6** double axis magnetic-field probes

Expanded set for MAXFEA-based studies and MEQ–EFIT comparison

36 flux loops + **36** double axis magnetic-field probes

NUMERICAL ASSESSMENT

Reconstruction focus: plasma-boundary recovery

EQUILIBRIUM STUDIES

FBT · LIUQE · FGS → Plasma geometry, global quantities and coil currents

CROSS-CODE COMPARISON

MAXFEA reference

MEQ and **EFIT** reconstructions

Boundary-gap agreement · Six cases

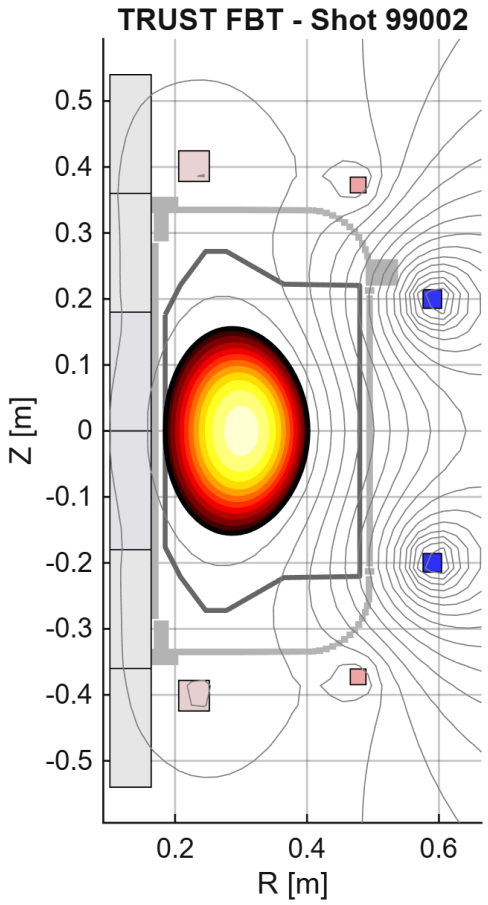
SCENARIO ASSESSMENT

METIS → Targets and reference boundary → **FBT** → Equilibria and coil currents

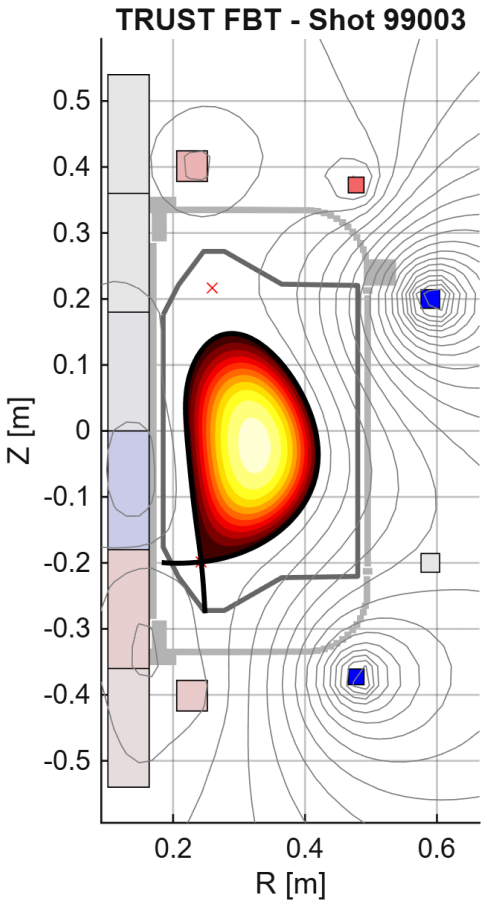
FBT EQUILIBRIUM DESIGN

Targets Plasma current: **110 kA** On-axis safety factor: **3** Poloidal beta: **0.1245**

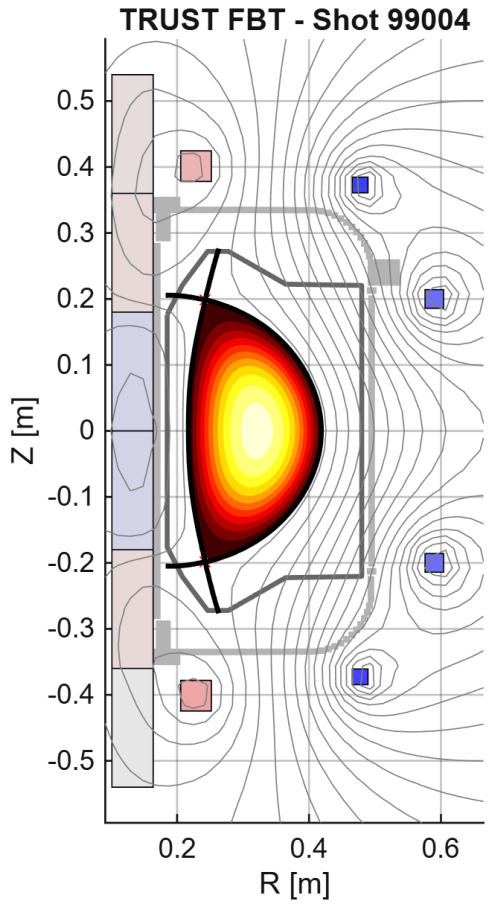
LIMITER



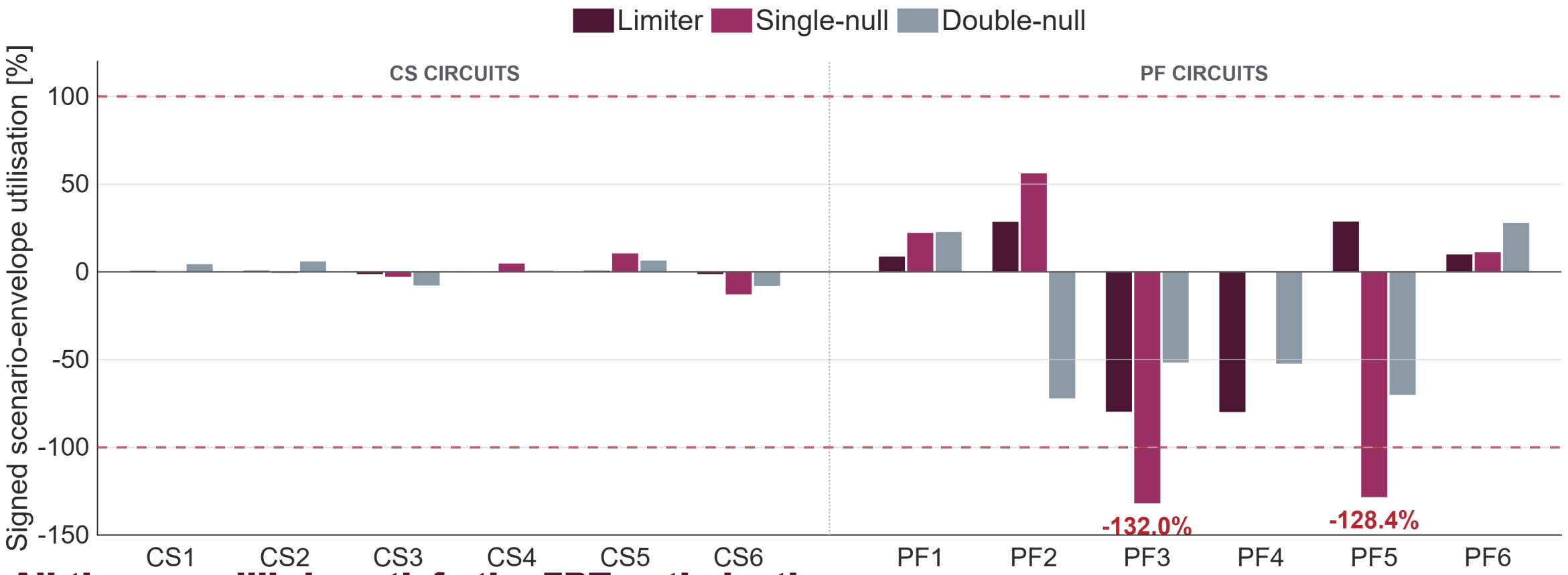
SINGLE-NULL



DOUBLE-NULL



FBT ACTIVE-CURRENT DEMAND



All three equilibria satisfy the FBT optimisation bounds.

PF3 and PF5 exceed the separate scenario envelope in the single-null case.

The scenario envelope was assessed separately and was not imposed as an FBT constraint.

LIUQE RECONSTRUCTION PATHWAYS

FBT reference → **Synthetic measurements** → **LIUQE**

Limiter, single-null and double-null

Reconstruction of native FBT equilibria

MAXFEA reference → **Synthetic measurements** → **LIUQE**

Single-null and nominal double-null

Direct reconstruction from MAXFEA data

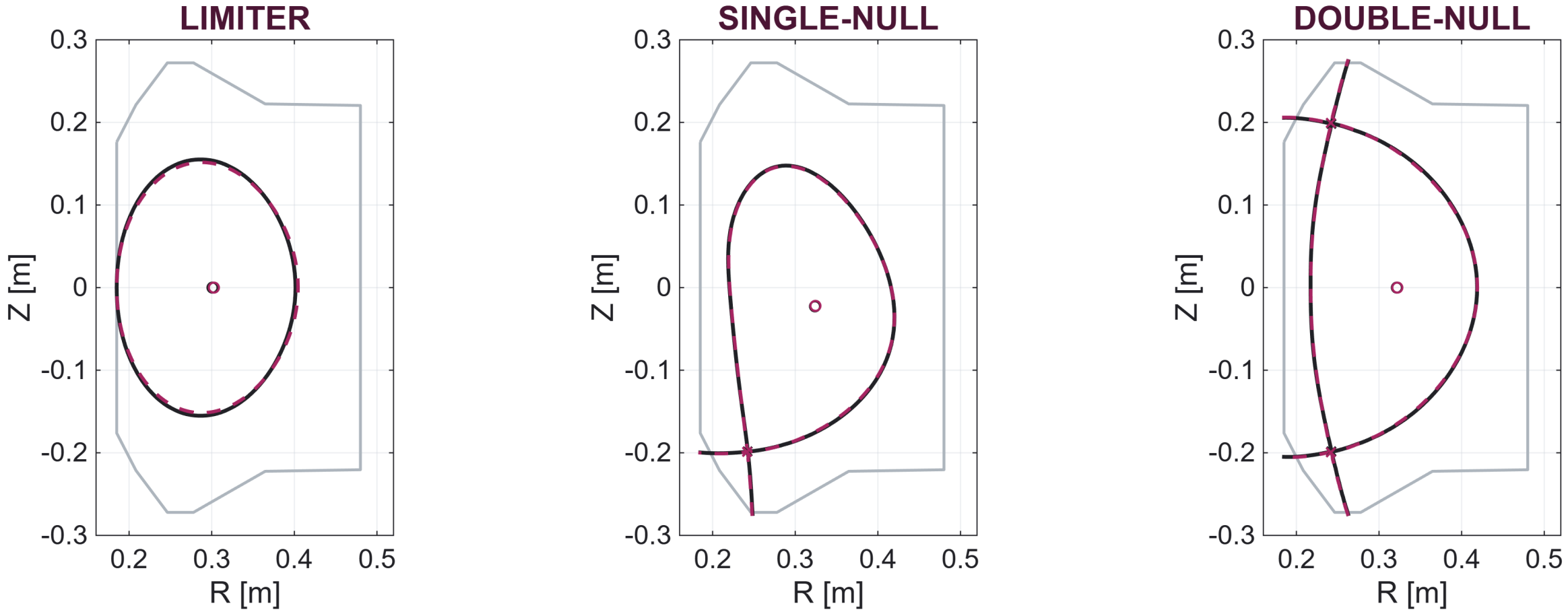
MAXFEA reference → **FBT** → **Synthetic measurements** → **LIUQE**

Single-null and nominal double-null

Separation of equilibrium transfer and reconstruction

Synthetic measurements with no added noise

RECONSTRUCTION OF FBT REFERENCES



Magnetic-axis radial shift
+1.486 mm

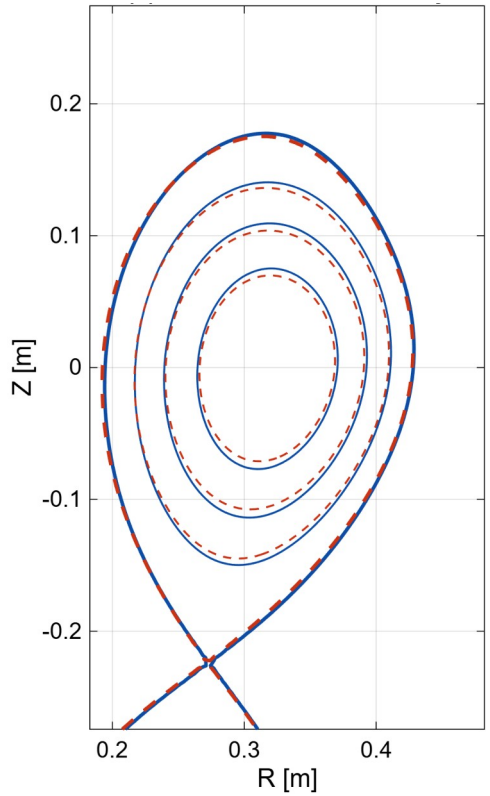
Lower X-point displacement
0.809 mm

Lower / upper X-point displacements
0.855 / 0.920 mm

DIRECT RECONSTRUCTION FROM MAXFEA

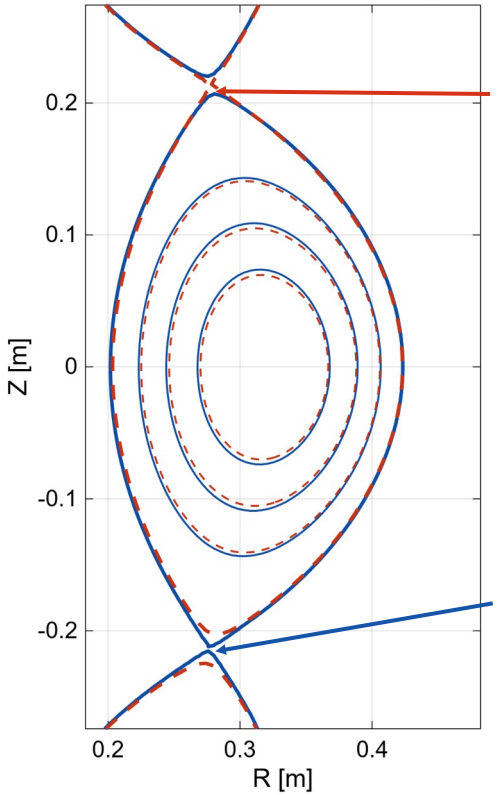
MAXFEA LIUQE

SINGLE-NULL



Lower X-point displacement: **1.377 mm**

DOUBLE-NULL



LIUQE boundary: upper null

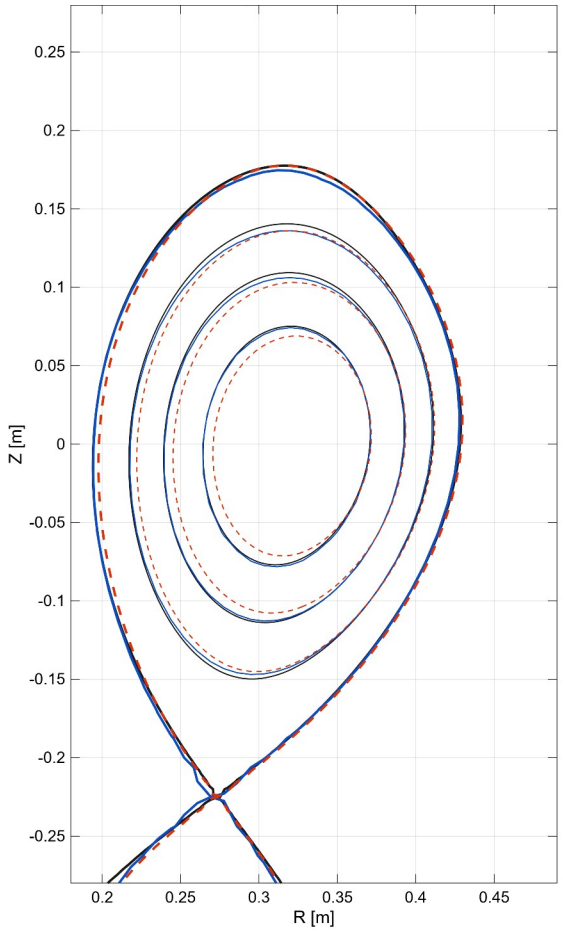
MAXFEA boundary: lower null

Lower X-point displacement: **1.539 mm**

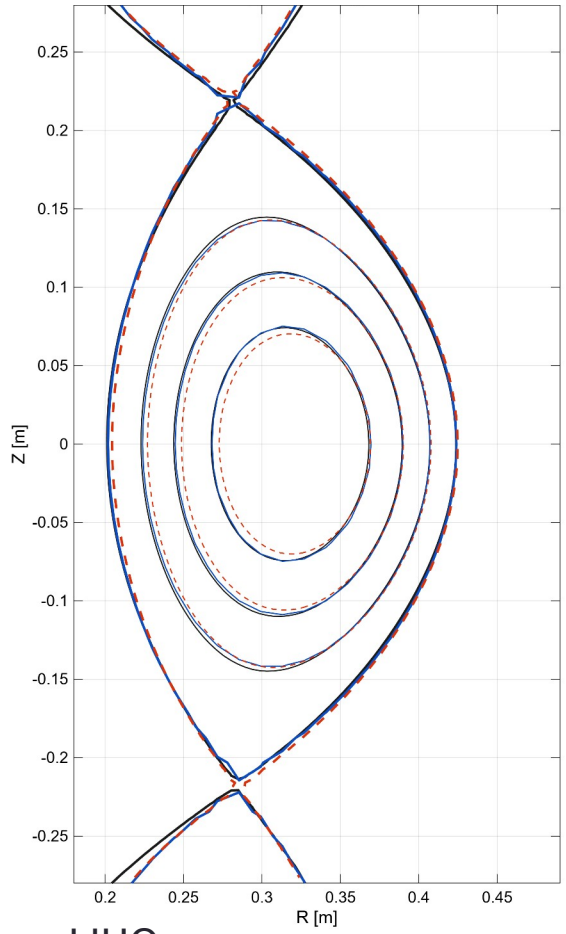
Close geometric agreement with MAXFEA in both reconstructions.

MAXFEA → FBT → LIUQE

SINGLE-NULL



DOUBLE-NULL



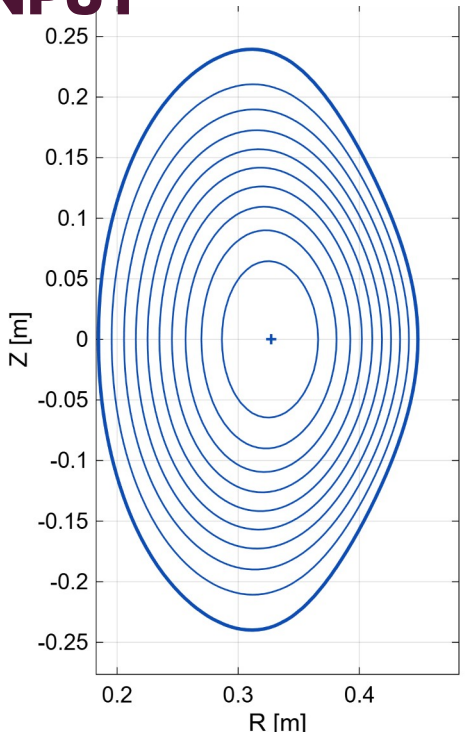
MAXFEA FBT LIUQE

Close boundary agreement across transfer and reconstruction in both configurations.

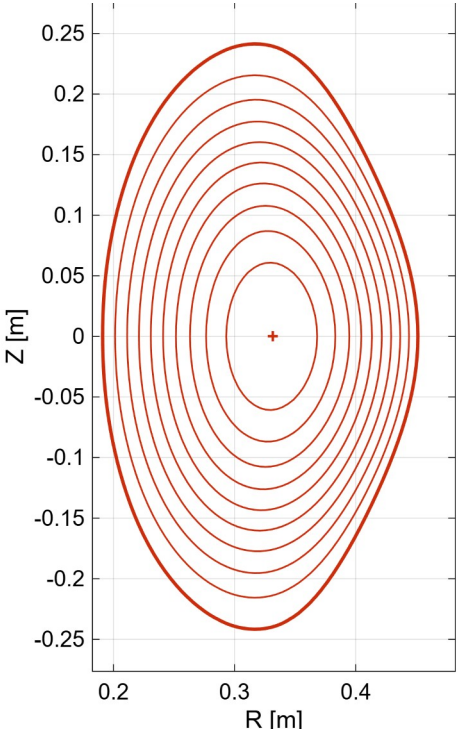
FGS WITH FBT DATA

Static forward solve with fixed active currents and prescribed I_p, q_A and β_p.

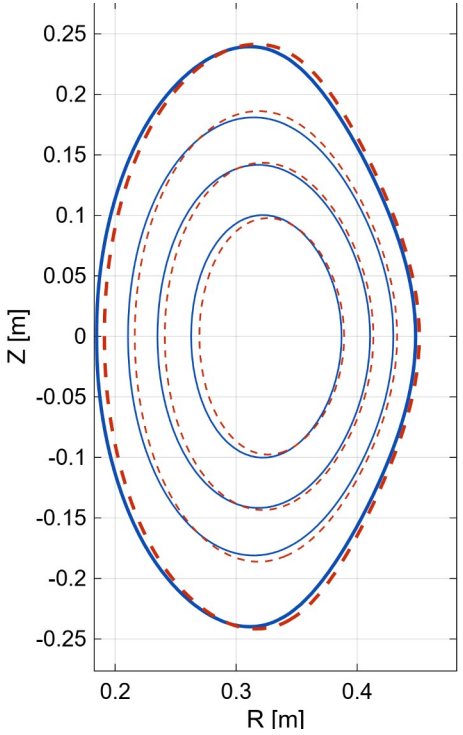
TRANSFERRED FBT INPUT



FGS RESULT



INPUT–FGS OVERLAY



Transferred input — — — FGS result

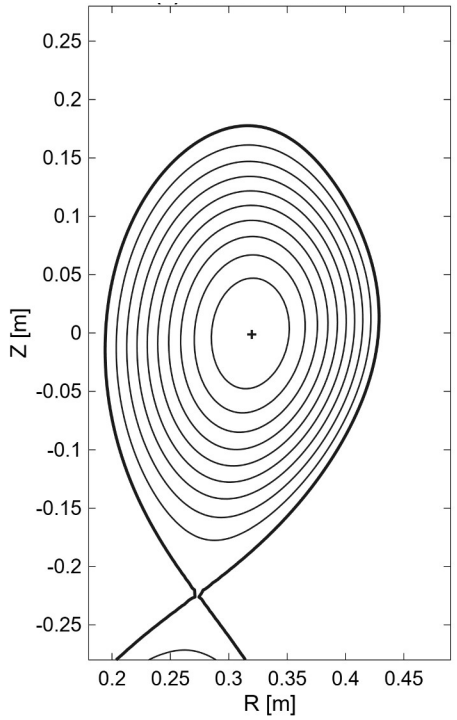
Magnetic-axis radial shift: **+4.306 mm**

FGS preserves the fixed currents and selected global constraints while the geometry adjusts.

FGS WITH MAXFEA DATA

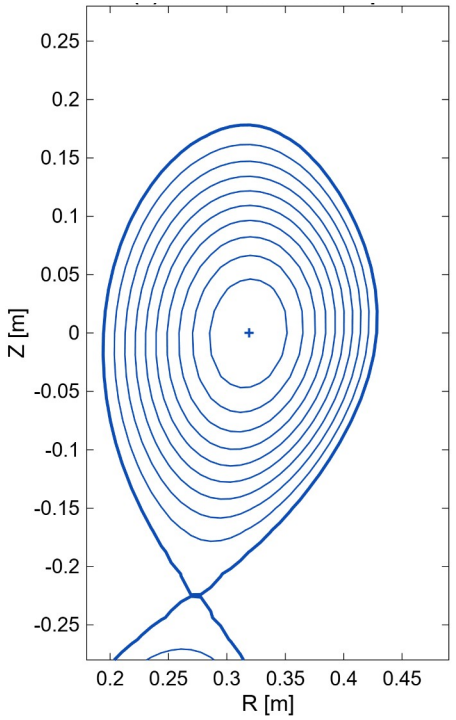
Comparison of the MAXFEA reference, transferred MEQ input and static forward result.

MAXFEA REFERENCE



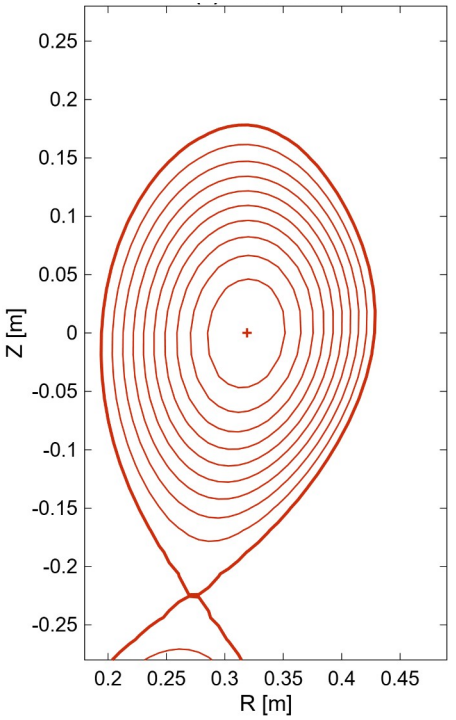
Plasma current, I_p MAXFEA: **110.000 kA**

TRANSFERRED INPUT



Transferred input: **110.753 kA**

FGS RESULT



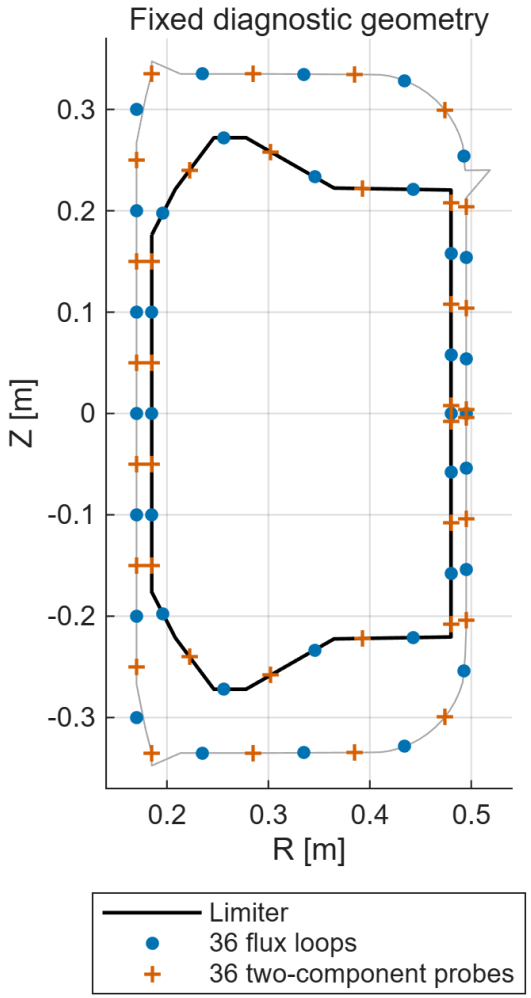
FGS: **110.753 kA**

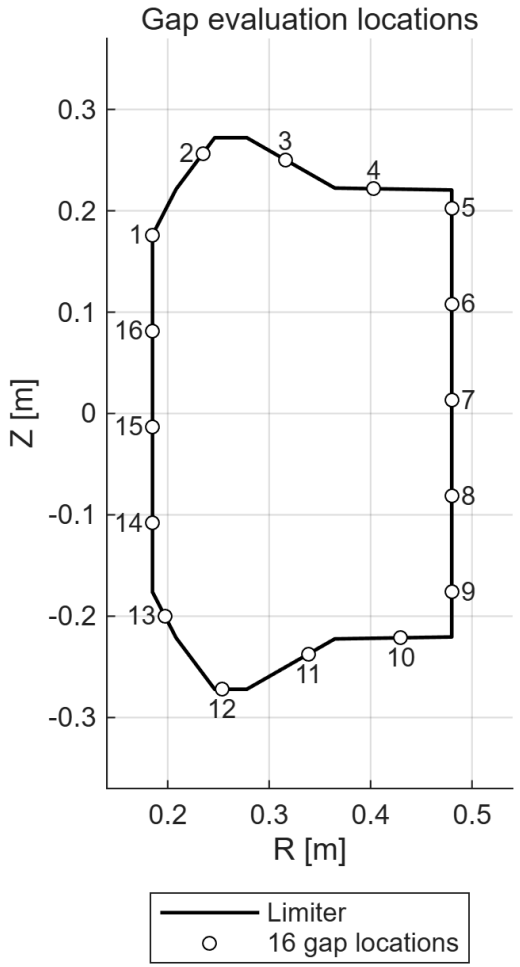
Differences arise during conversion; FGS retains the transferred equilibrium.

Active currents and the transferred I_p, q_A and β_p remain unchanged through the FGS solve.

MAXFEA–MEQ–EFIT COMPARISON

Six MAXFEA references reconstructed with MEQ and EFIT using noisy synthetic measurements.





SIX REFERENCE EQUILIBRIA

Configuration	Start of flat-top	End of flat-top
Double-null	DN SOF	DN EOF
Lower single-null	LSN SOF	LSN EOF
Upper single-null	USN SOF	USN EOF

RECONSTRUCTION INPUTS

36 flux loops + 36 double axis magnetic-field probes

108 magnetic observations per equilibrium

MEASUREMENT NOISE

Absolute and signal-dependent Gaussian contributions

Relative coefficient: 3%, plus an absolute noise floor

Boundary agreement is assessed against MAXFEA using 16 LCFS–limiter gaps.

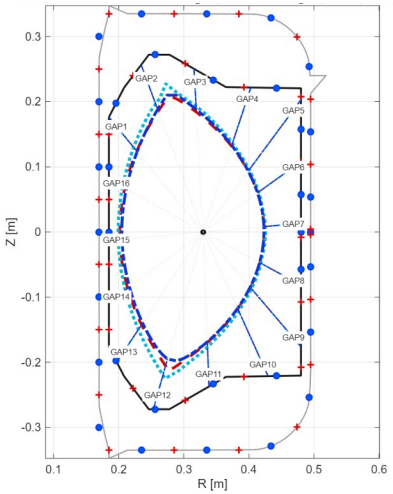
RECONSTRUCTED BOUNDARIES ACROSS SIX CASES

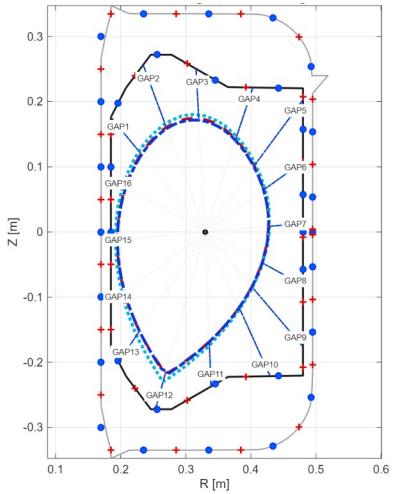
DOUBLE-NULL

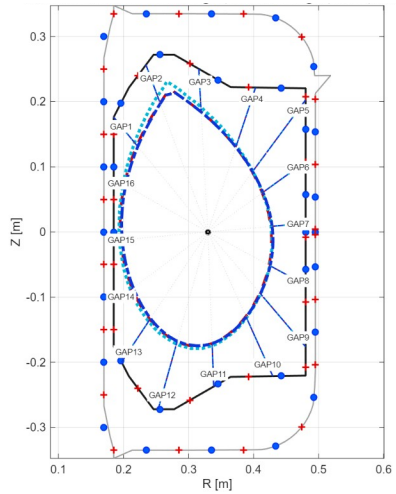
LOWER SINGLE-NULL

UPPER SINGLE-NULL

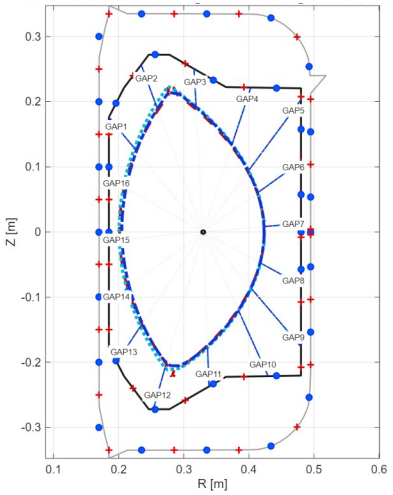
START OF FLAT-TOP

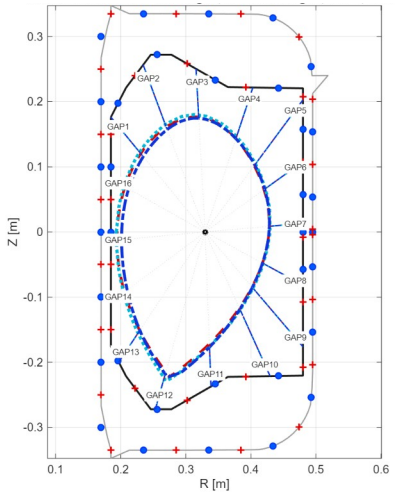


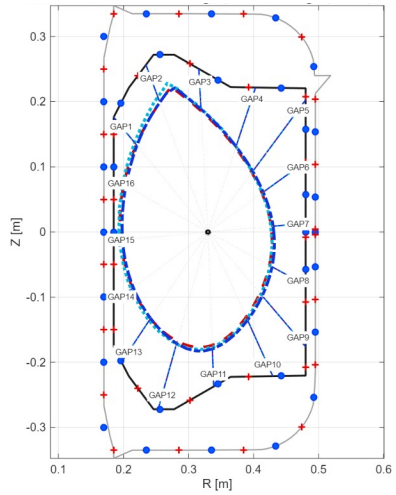




END OF FLAT-TOP



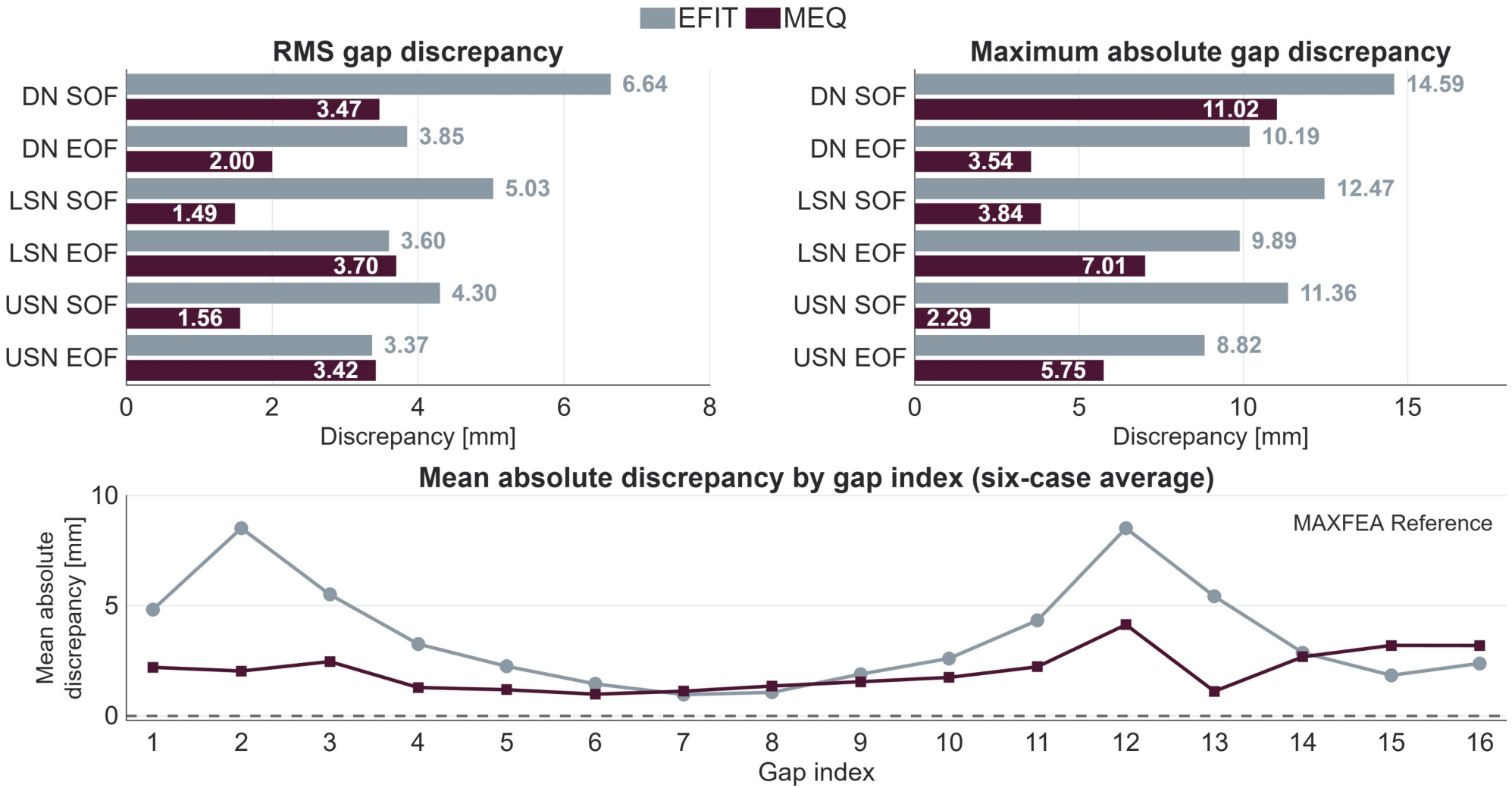




BOUNDARIES — MAXFEA — EFIT — MEQ

DIAGNOSTICS — Flux loops — B probes

BOUNDARY AGREEMENT ACROSS SIX CASES



MEQ has lower RMS in 4/6 cases and lower maximum discrepancies in all six.

METIS–FBT COUPLING

ONE-WAY COUPLING · METIS → FBT

METIS SCENARIO

Time-dependent plasma and geometry references.

REFERENCE INPUTS

Plasma targets: I_p, β_p and q_A.

Toroidal-field parameter: RB_t.

Reference LCFS: 64 (R, Z) points.

Reference currents for 12 active circuits.

The reference currents are optimisation targets, not fixed currents. The final FBT currents may differ from them.

FBT EQUILIBRIA

Inverse free-boundary equilibrium calculation.

Calculated plasma boundary.

Required currents for the 12 active circuits.

44 SAMPLED STATES · 10–438 ms

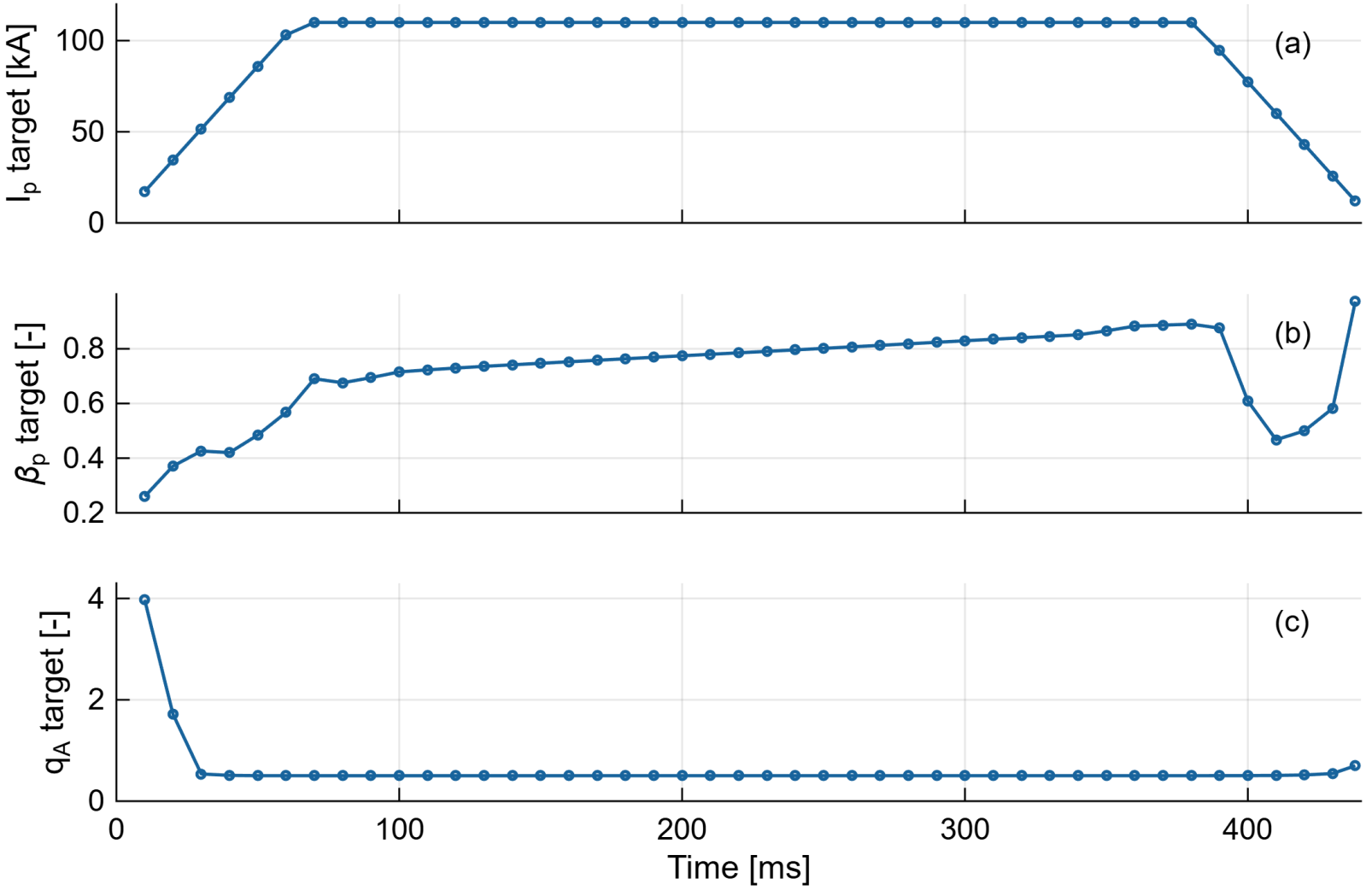
RAMP UP	FLAT-TOP	RAMP DOWN
10–60 ms	70–380 ms	390–438 ms
6 states	32 states · I_p target = 110 kA	6 states

FBT calculates the equilibrium and active-current demand at each sampled time.

Quasi-static sequence: circuit dynamics are not integrated, passive currents are zero, and voltage-limit checking is disabled.

TRACKING THE SCENARIO TARGETS

METIS REFERENCE TARGETS



FBT RESULTS

44/44

Sampled equilibria converged

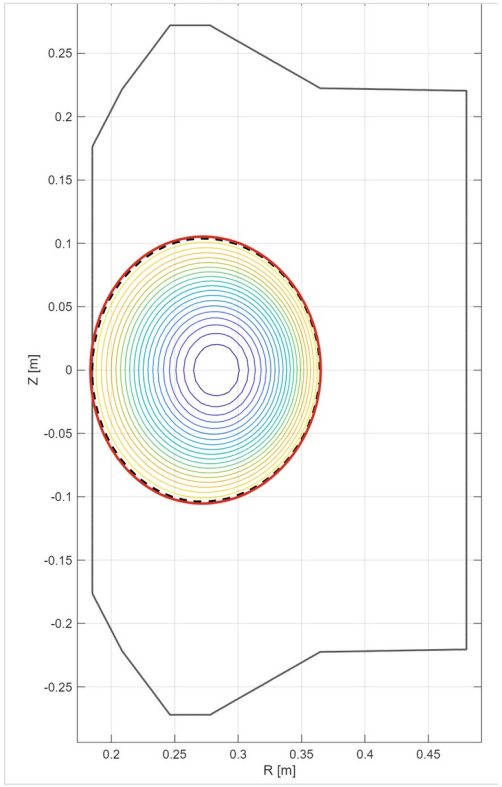
<0.01%

Maximum relative deviation from the three prescribed plasma targets

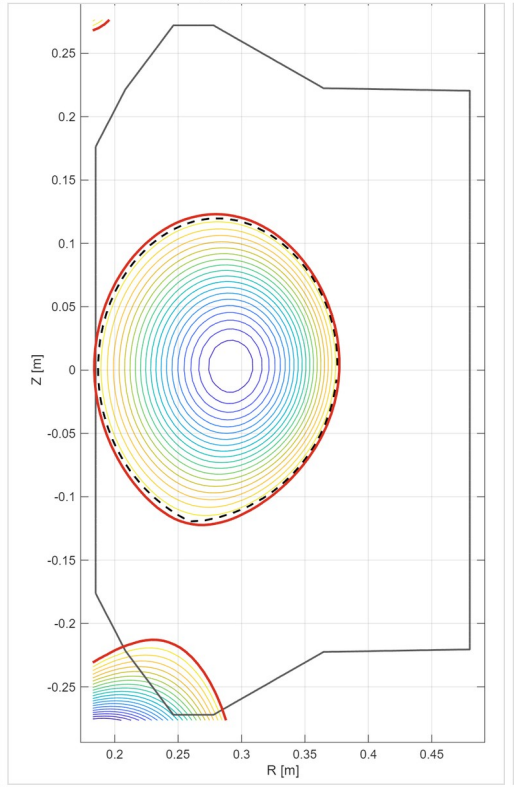
Across all 44 sampled equilibria

EARLY EQUILIBRIUM EVOLUTION

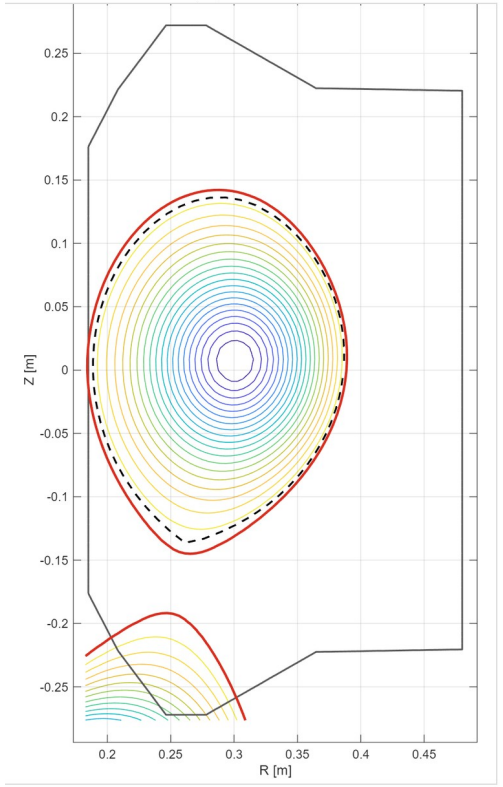
10 ms



20 ms



30 ms



TRUST first wall

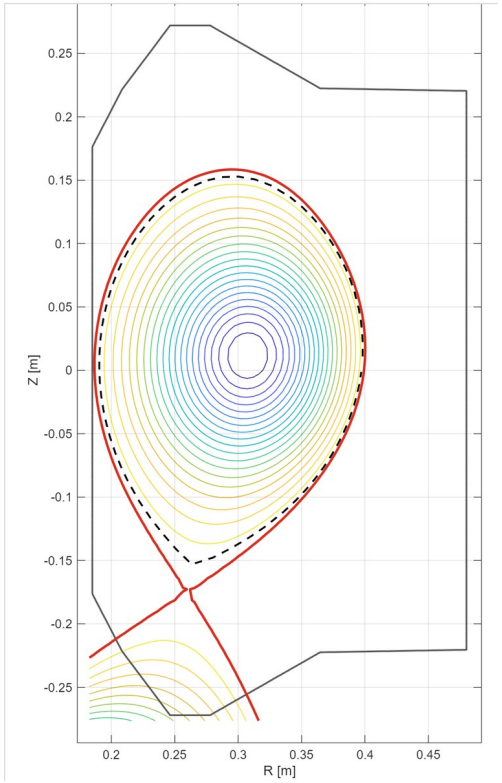
METIS reference LCFS

FBT boundary

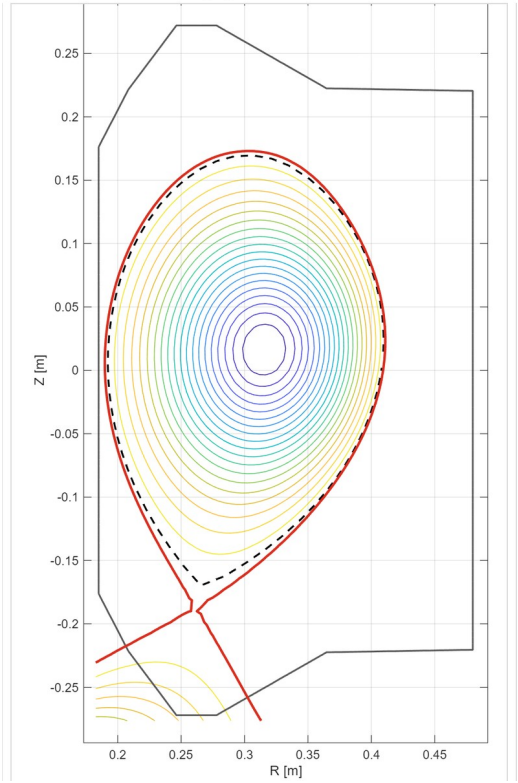
The plasma grows and elongates before lower single-null formation.

LOWER SINGLE-NULL FORMATION

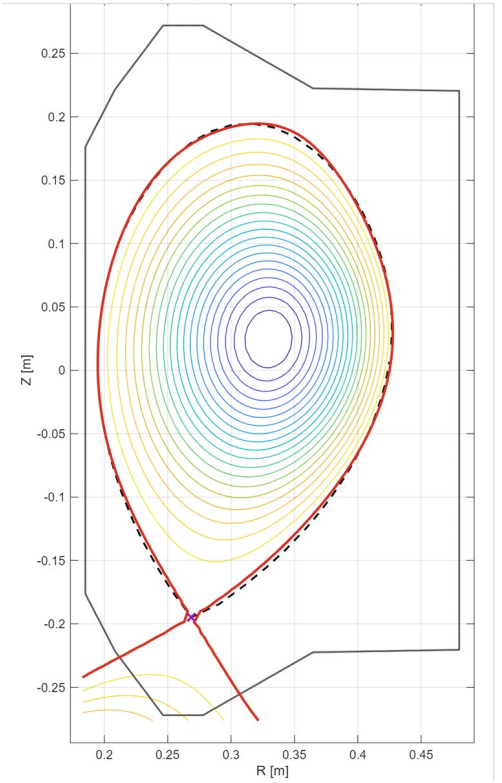
40 ms



50 ms



70 ms

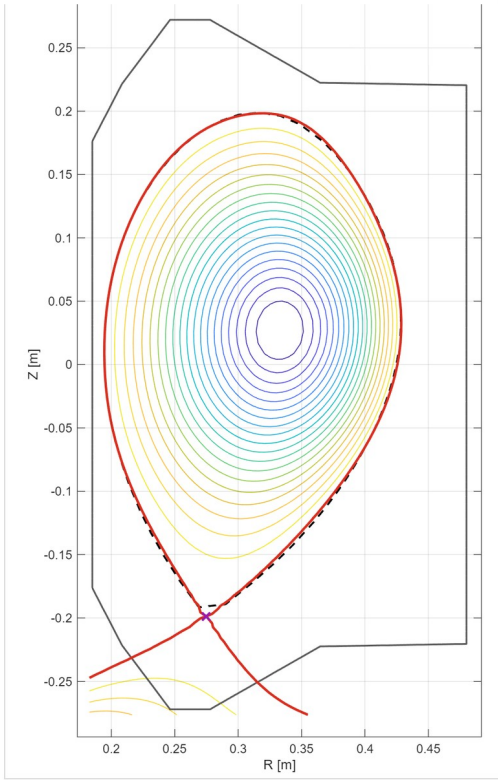


TRUST first wall — – – – METIS reference LCFS — FBT boundary × X-point

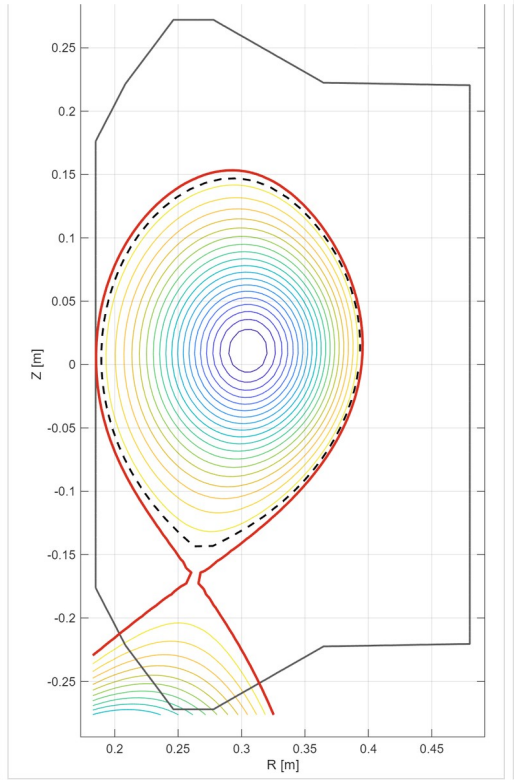
A lower X-point appears by 40 ms.

RETURN TO LIMITER CONFIGURATION

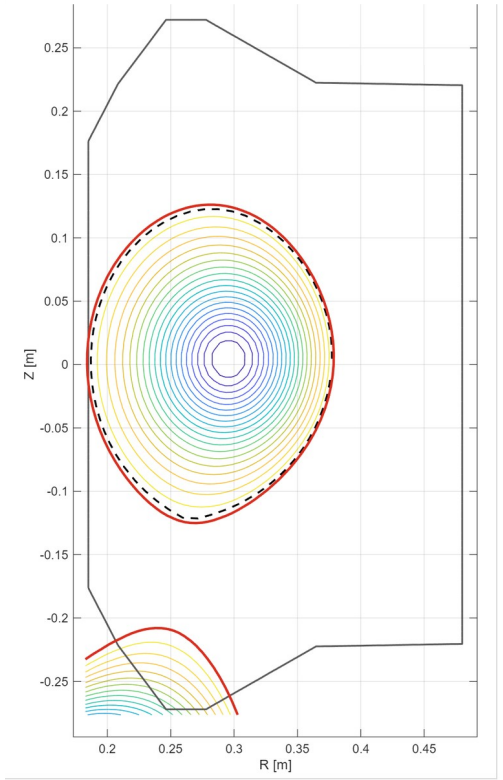
380 ms



410 ms



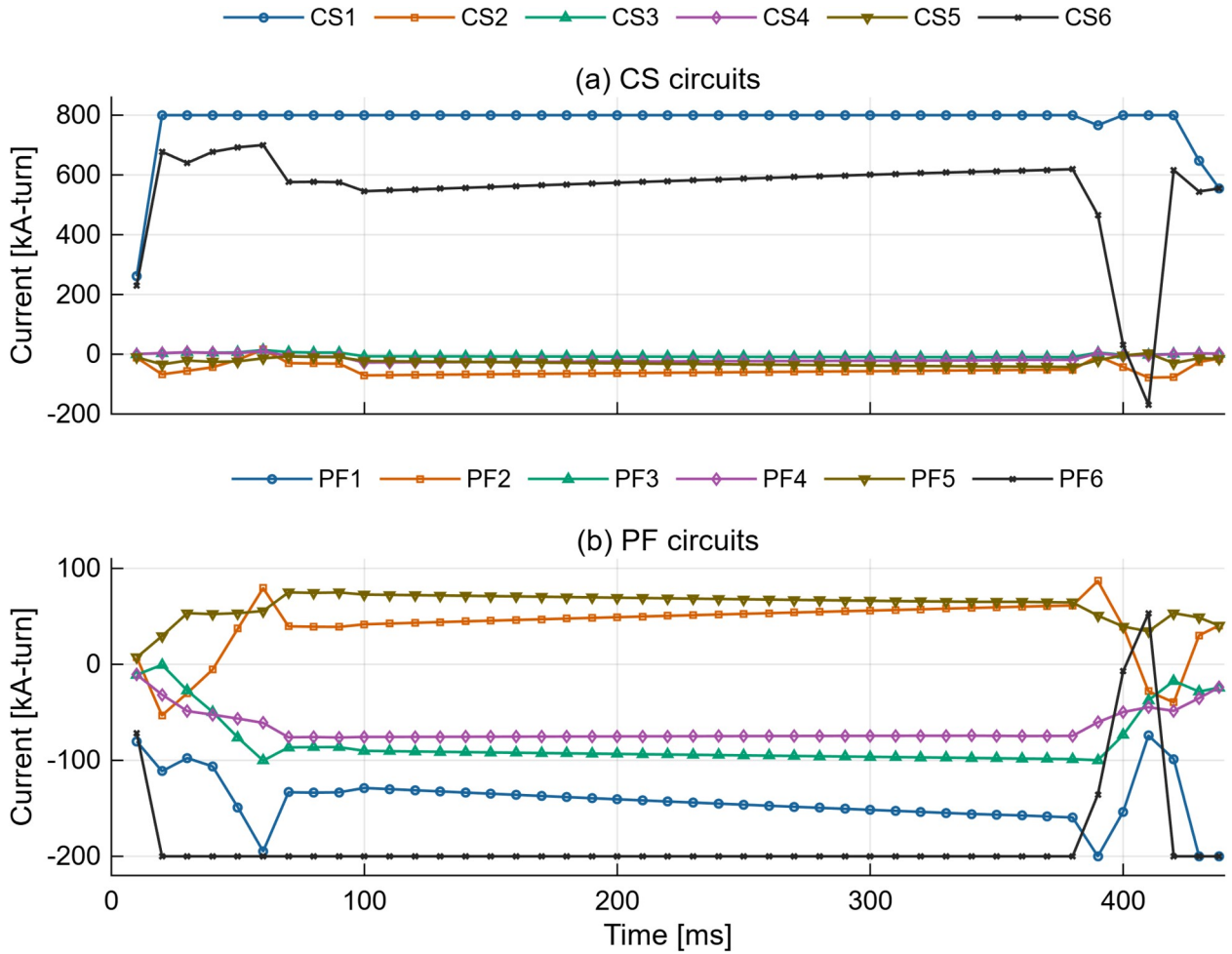
420 ms



TRUST first wall — — — METIS reference LCFS — FBT boundary × X-point

The lower X-point disappears between 410 and 420 ms.

SCENARIO CURRENT DEMAND



CURRENT-LIMIT ASSESSMENT

44/44

States within FBT optimisation bounds.

FBT bounds: CS ±800; PF ±200 kA-turn.

The coupling quantifies coil-current demand along the reference scenario.

CONCLUSIONS AND FUTURE DEVELOPMENTS

MEQ adapted to TRUST for equilibrium design, reconstruction and scenario assessment.

MAIN FINDINGS

Plasma-boundary reconstruction

Boundary recovery demonstrated across the tested limiter, single-null and double-null configurations.

Six-case cross-code comparison

MEQ RMS boundary-gap discrepancies:

1.49–3.70 mm

Six MAXFEA comparison cases · 16 sampled boundary gaps.

Lower RMS than EFIT in 4/6 cases and lower maximum absolute discrepancies in 6/6.

NEXT STEPS

Experimental validation

Assess reconstructions against calibrated TRUST measurements when available.

Broader cross-code assessment

Extend comparisons to other equilibrium codes and reference cases.

Scenario current constrains

Impose the reference scenario current envelope as a constraint in FBT inverse equilibrium calculations.

Thank you

for your attention